Huawei Access Node JAY-ACC-68 Optical Transport Failure SOP (ALARM_7781)

Node: JAY-ACC-68 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_7781

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Node JAY-ACC-68 at Jayanagar Metro Site is a critical failure domain that requires immediate attention. This failure is attributed to the degradation of transceiver modules (SFP/XFP) or fiber cuts, leading to a loss of signal power at the receiver end. The impact of this failure is a Minor (ALARM_7781) alarm, which necessitates a thorough investigation and resolution to prevent network outages. The Huawei Access Node JAY-ACC-68 utilizes advanced Optical Transport Network (OTN) protocols, including the Generalized Multi-Protocol Label Switching (GMPLS) and the Synchronous Digital Hierarchy (SDH) for signal transport. The OTN protocol stack, coupled with the SFP/XFP transceivers, enables high-speed data transmission over optical fibers. However, the sensitivity of these transceivers to environmental factors, such as temperature and humidity, can lead to signal degradation and eventual failure. Moreover, fiber cuts or damaged fibers can also cause signal loss, resulting in the Optical Transport Failure. Therefore, it is essential to promptly identify and resolve this failure to maintain network reliability and uptime.

Pre-requisite Checks

- Verify the site location and ensure remote hands are dispatched to Jayanagar Metro Site.
- Check the redundant link status to ensure that it is operational and can take over in case of a failure.
- Identify the affected optical interface and its corresponding transceiver module (SFP/XFP).

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP transceiver module	High
Fiber cuts or damaged fibers	Medium
Rx/Tx power loss due to environmental factors	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical interface interface-name, Expected Output: Interface status, Rx/Tx power levels, and SFP/XFP temperature, Description: This command displays the status of the optical interface, including Rx/Tx power levels and SFP/XFP temperature, which can help identify signal degradation or fiber cuts.

Command: display optical transceiver transceiver-name, Expected Output: Transceiver status, temperature, and optical signal quality, Description: This command displays the status of the transceiver
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module, including temperature and optical signal quality, which can
help diagnose SFP/XFP degradation.
```

Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP transceiver module, Command: replace optical transceiver transceiver-name, Description: Replace the degraded SFP/XFP transceiver module with a new one to resolve the signal degradation issue.

Step 2: Step: Repair or replace the damaged fiber, Command: replace optical fiber fiber-name, Description: Repair or replace the damaged fiber to resolve the signal loss issue.

Post-Check Verification

```
Command: display optical interface interface-name, Expected Output:
Interface status, Rx/Tx power levels, and SFP/XFP temperature,
Description: Verify that the optical interface is operational and
Rx/Tx power levels are within normal ranges.

Command: display optical transceiver transceiver-name, Expected
Output: Transceiver status, temperature, and optical signal quality,
Description: Verify that the transceiver module is operational and
optical signal quality is within normal ranges.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver module, Command: restore optical transceiver transceiver-name, Description: Restore the original SFP/XFP transceiver module if the replacement fails to resolve the issue.

- Step: Restore the original fiber, Command: restore optical fiber fiber-name, Description: Restore the original fiber if the replacement fails to resolve the issue.